

6502

50th Anniversary Computer Badge
(1975–2025)

A retro computer in badge format for 8-bit 6502 fans

By Frank Eggen 28/12/2025 V2.6

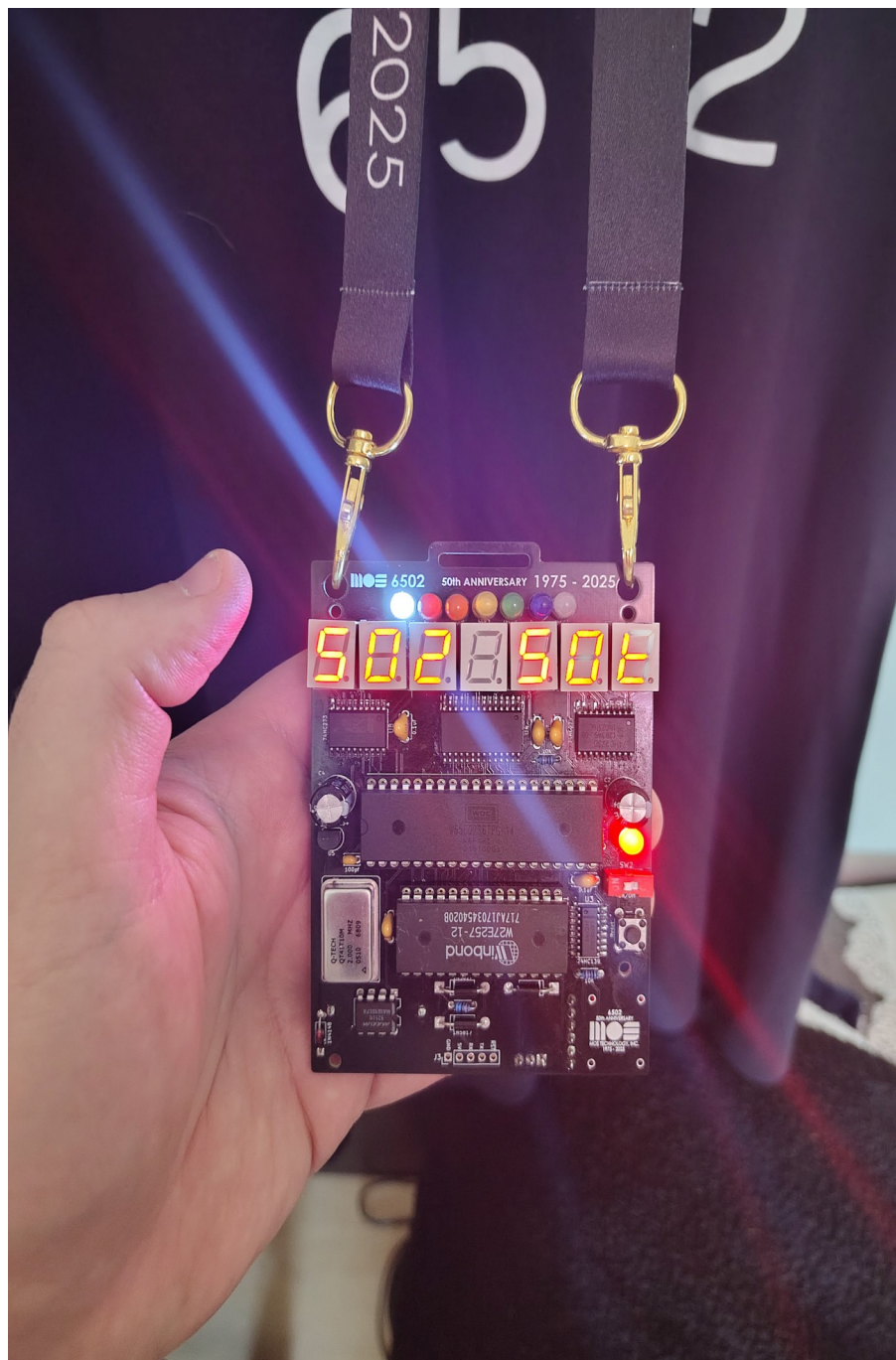
Based on an idea by **Lee Hart, Daryl Rictor, and Josh Bensadon** (revised and updated).

👉 Original project: sunrise-ev.com/6502.htm

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50 Years of MOS 6502



The legendary **MOS 6502** was introduced in 1975 and changed the world of computing. It delivered performance at a low price and paved the way for the triumph of **home computers**.

Well-known systems with 6502:

- Apple II
- Commodore VIC-20 & C64
- Nintendo NES
- BBC Micro
- Atari VCS

👉 More about the history: team6502.org

What is the Computer Badge?

The **MOS 6502 50th Anniversary Computer Badge** is a **minimalist reissue** of an 8-bit microcomputer with a 6502 CPU in badge format – easy to replicate with today's components.

Highlights

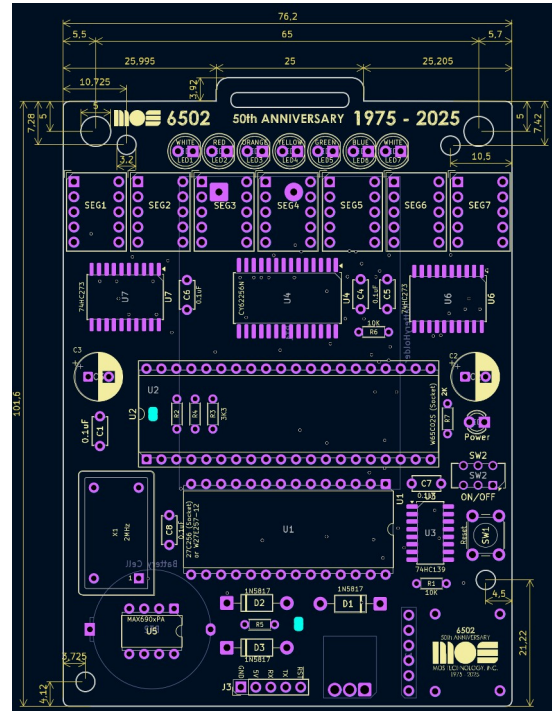
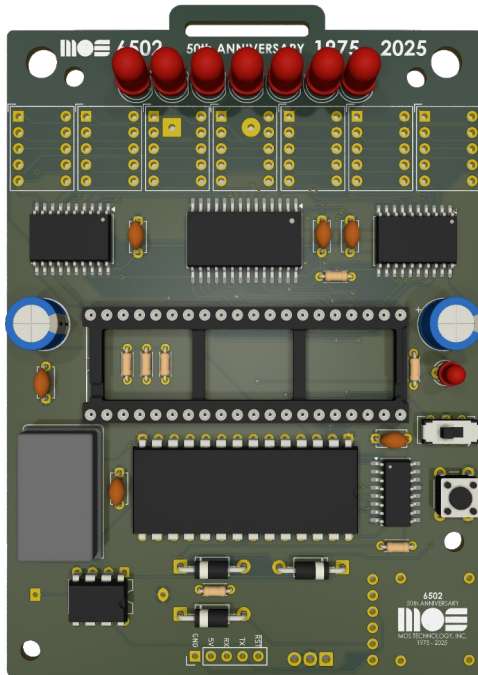
- **W65C02S microprocessor @ 2 MHz**
- **32 KB RAM** with battery backup
- **32 KB EPROM** with floating point BASIC + 6502 monitor
- **7-digit 7-segment LED display + 7 LEDs** for effects
- **RS-232-compatible interface** for data exchange
- **USB-C (CH340) serial TTL interface** – can be used directly on a PC
- **DC/DC step-up module** for stable battery operation
- **CR2032 battery** for SRAM data retention

Fully functional, programmable in **Assembler** or **BASIC** – and still small enough to be worn as a retro name tag.

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Variants

Badge version



- 7.62 × 10.16 cm (3 × 4 inches) – classic badge format
- PCB layout (Gerber) → <https://github.com/FraEgg/The-MOS-6502-50th-Anniversary-Computer-Badge/tree/2.6/gerber>

What can you use it for?

- As a **retro name tag** with scrolling text of your choice
- To **celebrate 50 years** of the **6502** and the home computer era
- As a **teaching project** for assembler, BASIC, and hardware nostalgia
- For **tinkering and having fun** with a minimal number of parts
- As a **cool collector's item** for the anniversary
-

Technical specifications

- W65C02S @ 2 MHz
- 32 KB RAM with battery backup
- 32 KB EPROM (BASIC + monitor)
- 2× 8-bit output latches
- 7-digit 7-segment LED display + 7 LEDs
- Serial Soft RS-232 interface, 9600 8N1
- USB-C to TTL CH340 UART adapter (BTE17-06B)
- DC/DC 5 V step-up boost converter
- Charging function for 3× AA NiCd batteries

👉 This gives you a **fully-fledged retro system** – minimalist, educational, and a real eye-catcher at any nerd gathering. 😊

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Bill of materials (BOM)

Created on 2025-10-06

Note: A DIP socket is recommended for **U1** and **U2**.

Pos.	Ref	Value	Qty	Description	Package	Datasheet
1	U1	27C256 or W27E257-12 (Socket)	1	EPROM 256 KiBit (Socket)	DIP-28_W15.24mm	Link
2	U2	W65C02S (Socket)	1	W65C02S 8-bit Microprocessor Dual 2-to-4 line	DIP-40_W15.24mm_Socket	Link
3	U3	74HC139	1	decoder/demultiple xer	SOP-16_4.4x10.4mm_P1.27mm	Link
4	U4	CY62256N	1	256-Kbit (32 K × 8) Static RAM	SOP-28_8.4x18.16mm_P1.27mm	Link
5	U5	MAX690xPA	1	IC SUPERVISOR 1 CHANNEL 8DIP	DIP-8_W7.62mm	Link
6	U6,U7	74HC273	2	IC FF D-TYPE SINGLE 8BIT 20SOIC	SOIC-20W_7.5x12.8mm_P1.27mm	Link
7	R1,R6	10K	2	Resistors 10K 1/4W 1% Metal Film	Axial	Link
8	R2-R4	3K3	3	Resistors 3K3 1/4W 1% Metal Film	Axial	Link
9	R5	10R	1	Resistors 10R 1/4W 1% Metal Film	Axial	Link
10	R7	2K	1	Resistors 2K 1/4W 1%	Axial	Link
11	C1,C4- C8	0.1uF	6	CAP CER 0.1UF 50V Z5U RADIAL	C_Disc_D4.7mm_W2.5mm	Link
12	C2,C3	470uF	2	Electrolytic Caps LOW IMPEDANCE	CP_Radial_D8.0mm_P3.50mm	Link
13	D1-D3	1N5817	3	20V 1A Schottky Barrier Rectifier	D_DO-41_SOD81	Link
14	X1	2MHz	1	XTAL OSC XO 2.0000MHZ HCMOS TTL	Oscillator_DIP-14	Link
15	LED1,7	WHITE	2	White LED Indication 2.3V	LED_D5.0mm	Link
16	LED2	RED	1	Red LED Indication 2.3V	LED_D5.0mm	Link
17	LED3	ORANGE	1	Orange LED Indication 2.3V	LED_D5.0mm	Link
18	LED4	YELLOW	1	Yellow LED Indication 2.3V	LED_D5.0mm	Link
19	LED5	GREEN	1	Green LED Indication 2.3V	LED_D5.0mm	Link
20	LED6	BLUE	1	Blue LED Indication 2.3V	LED_D5.0mm	Link
21	LED8	Power	1	Red LED Indication (Power) 2.3V	LED_D3.0mm	Link
22	SEG1-7	SC39-11EWA	7	7-segment display, com. cathode	Sx39-1xxxxx	Link
23	SW1	Reset	1	SWITCH TACTILE	SW_PUSH_6mm	Link

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Pos.	Ref	Value	Qty	Description	Package	Datasheet
				SPST-NO		
24	SW2	ON/OFF	1	SWITCH SLIDE	JS202011CQN	Link
				DPDT 300MA 6V		
25	J1	5V DC-DC	1	0.9-5V To 5V Step-Up Module	THT	Link
26	J2	CH340N	1	CH340N SOP8	USB-to-TTL-Modul	Link
				USB to TTL module		
27	J3	Header	1	Generic connector, 01x05	PinHeader 1x05	-
28	JP1	*WD/MOS	1	Solder Jumper, 2-pole, open	-	-
29	JP2	Charging	1	Solder Jumper, 2-pole, closed	-	-
				Holder 3x AA		
30	BT1	3x AA	1	(Warning: NiCd only!)	Keystone_3xAA	Link
31	BT2	CR2032	1	Holder CR2032 (Backup)	Panasonic_CR2032	Link

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Assembly

First check the parts list and make sure you have all the parts. For assembly, you will need a **soldering iron**, **solder**, and **flux** (for the SMD ICs). **Soldering tweezers**, a **desoldering pump**, and **desoldering braid** are helpful.

Recommended order:

1. **Solder SMD ICs:** U3 (74HC139), U6/U7 (74HC273), then U4 (SRAM).
Tip: Apply a little flux to the **first pin at the top left** and the **last pin at the bottom right**, align the IC, and fix these two pins with a little solder. Once everything is in place, solder the remaining pins one after the other. **Make sure the polarity/pin 1 marking is correct.**
2. **Populate the sockets:** U1 (EPROM) and U2 (W65C02S).
3. Solder in **U5 (MAX690 or MAX692).**
4. Insert **X1 (2 MHz oscillator).**
5. **Diodes:** Fit **D1, D2 and D3 (1N5817).**
6. Fit **resistors R1–R7.**
7. Solder **ceramic capacitors C1, C4–C8.**
8. Solder **electrolytic capacitors C2/C3 (470 μ F, 10–16 V).**
9. Insert **pushbutton SW1 and DIP switch SW2 (ON/OFF).**
10. **LEDs:** Install LEDs 1–7 (LED strip) and **LED 8 (power LED).** **Note the polarity!** The hole with the **square** solder pad is the **cathode**.
11. **J1 (DC/DC step-up 5 V, 10.5×11):** Mount with **3-pin headers** on the **back**. **Observe polarity.** The coil of the module points **away from the board**.
Not using a step-up? Then bridge **pins 1 and 3** of J1 (**do not** connect to pin 2). This is only recommended for stable 5 V (laboratory power supply/USB-C power bank).
12. Connect **J2 (CH340N USB-to-TTL module)** to the **5-pin header** on the back (note the polarity).
13. **Cleaning:** Remove flux residues (isopropanol).
14. **7-segment displays:** now insert SEG1–SEG7. **Observe polarity.** There is a **dot** at the bottom right – align it **downwards** on the **front**.
15. Install **battery holders BT1/BT2** if necessary, **but do not** insert batteries/rechargeable batteries **yet**.

***Caution:** Only charge 3× AA NiCd rechargeable batteries (1.2 V)! The charging electronics are **very simple**. **Do not** use Li-Ion or NiMH batteries → **risk of fire/explosion**. **Deactivate** the charging function for **batteries (open JP2)**.*
16. **Finally**, insert U2 (W65C02S CPU) and the programmed U1 (EPROM) into the sockets.

Your **MOS 6502 50th Anniversary Computer Badge** is now complete. 🎉

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Jumper options

The **badge** has **three jumpers** for different configuration options.

1. **JP1 – Processor type**
Open = default for the **W65C02S** (Western Design Center). For older types (MOS, Rockwell, UMC, Synertek), **GND** must be connected → **close JP1**. Use **2 MHz variants** (e.g., MOS **6502AD**, Rockwell **R6502AP**, UMC **UM6502A**). The circuit has only been tested with **W65C02S**. CMOS variants (**65C02**) have significantly lower power consumption. **BadgeOS** uses extended **CMOS opcodes** – the original **NMOS variant** requires its own OS.
2. **JP2 – Charging the AA NiCd batteries**
Closed by default. If you use **3× AA 1.5V batteries** instead of NiCd rechargeable batteries, **JP2 must be opened** so that no charging voltage is applied (for external power supply via USB-C J2 or pin header J3).

First start: Let it run!

1. **SW2 to OFF.**
2. Connect **5V** to **J2 (USB-C/CH340)** or **J3 (pin header)** (note the polarity!).
3. **SW2 to ON.** The power LED lights up. You should see the scrolling text "6502 badge for VCF" or "6502 50th Birthday 1975-2025" on the 7-segment display.

Test battery operation:

SW2 **OFF**, disconnect external 5 V. Insert **CR2032** (for SRAM backup) and **3× AA NiCd**. In pure battery mode, the power LED **does not** light up as long as SW2 is **OFF**. **SW2 ON** → the computer starts up, the scrolling text appears. Charging time for NiCd batteries via USB-C: approx. **6–8 hours**. Runtime: **6–12 hours** (depending on LED series resistors/brightness).
The circuit board has four **mounting holes** (e.g., for spacers or housing).

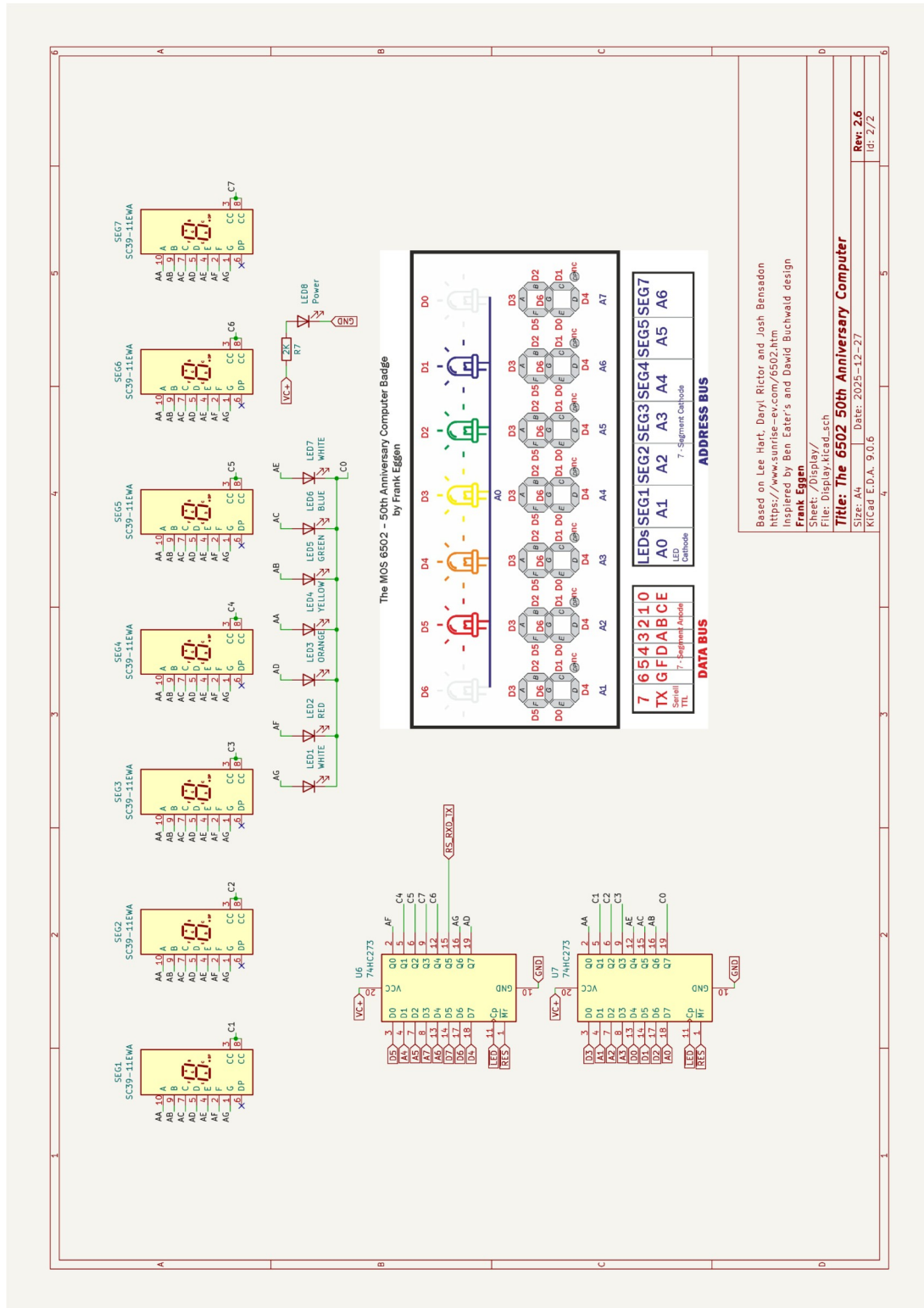


An additional highlight is the **lanyard**, which can be attached to the upper holes – so you can comfortably wear the **badge** around your neck.



→ <https://github.com/FraEgg/The-MOS-6502-50th-Anniversary-Computer-Badge/blob/2.2/downloads/Schematic.pdf> (PDF, ~0.2 MB)





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The **badge computer** is a complete minimalist computer: **6502 CPU with 2 MHz, 32 KB RAM (U4), and 32 KB ROM (U1)**. **U3** forms the address logic (chip select RAM/ROM) and controls the LED segments. The two **74HC273 (U6/U7)** drive the LEDs and the 7-segment display as latches/flip-flops. **Bit 7** serves as **TX** for the serial interface.

RX is read in via **IRQ (pin 4)** of the CPU. This is a **software-based RS-232 interface** (soft UART); the **OS in the ROM** handles the timing/handling (9600 baud, 8N1). The **CH340** module at **J2** establishes the USB connection (install drivers under Windows if necessary). You can communicate via **9600,8N1** using **PutTTY/Minicom**.

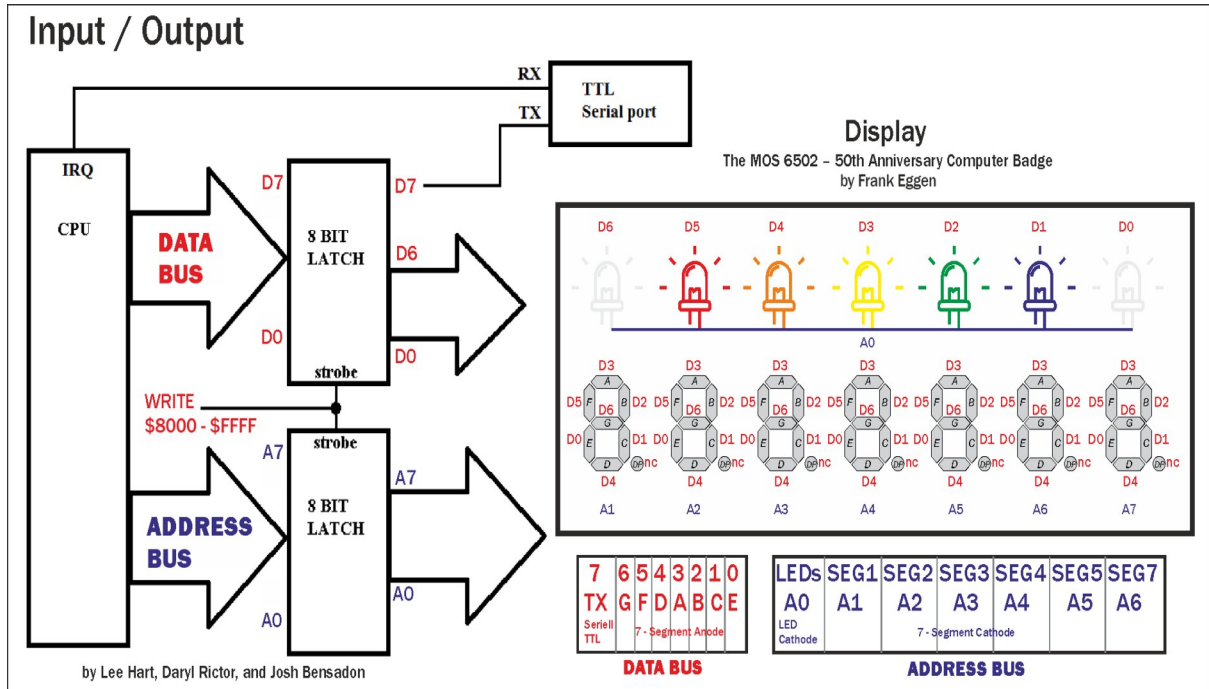
Memory

The badge has **32 KB RAM** from address **\$0000–\$7FFF** and **32 KB ROM \$8000–\$FFFF**.

Start	End	Range
8000h	FFFFh	ROM 32 KB (EPROM U1)
0000h	7FFFh	RAM 32 KB (SRAM U4)

Input/output

The two **74HC273 8-bit latches (U6/U7)** control the input/output. A common clock is derived from a **write pulse (R/W)** in the ROM address space. All **8 data bus** and the **low 8 address bus lines** are latched as soon as writing occurs in the range **\$8000–\$FFFF**. These latches drive the **8 rows** and **7 columns** of **multiplexed LEDs** and 7-segment displays. For layout reasons, data/address pins are mixed between the latches; logically, each line is correctly assigned (**D0 → D0, A4 → A4, ...**).



Input-output schematics

This "unusual" design fulfills conflicting goals:

- All **15 LED output bits** can be updated simultaneously with a **single** instruction (flicker-free)

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- display).
- **Minimal hardware** (no separate UART).
- **Simple/small PCB layout**.

Bit 7 of the data latch is not needed for the LEDs and therefore serves as **TX**. The CPU's **IRQ** is the RX input. Together with software, this results in a **minimalist interrupt-controlled** serial port (monitor + BASIC run on it).

LED display

The **data bus latches** drive the **LED anodes (segments)**, the **address bus latches** drive the **LED cathodes (digits)**.

Data bits 0–6 select segments **A–G**; **data bit 7** is **TxD**.

Address bits 1–7 select the **7 digits**; **address bit 0** selects the **7 discrete LEDs**.

Anodes are **active high**, cathodes **active low**. Example: Switch on segment **A** of the **leftmost digit** → **data bit 3 = 1**, **address bit 1 = 0** → write the value **\$08** to address **\$80FD**. (Any address $\geq$ **\$8000** has the same effect.)

***Note on the firmware:** Compared to the original by Lee Hart/Daryl Rictor/Josh Bensadon, this version uses **7 LEDs** on the data bus **instead of 5**. The original firmware only controls 5 LEDs. Therefore, use the modified firmware **BadgeOS_50th.zip** (see **Downloads**), which uses all 7 LEDs. In the **basic/** folder, you will find examples of LED running lights.*

Serial port

The serial port is a **TTL asynchronous** data transfer (TxD/RxD/GND). It is **similar to RS-232**, but uses 5 V/0 V levels and **non-inverted** data (idle = 5 V = 1). You can either

- use a **TTL-to-RS-232 adapter** on **J3** or
- plug a **CH340N USB-to-TTL module** into **J2** (recommended).

The configuration is **9600, N, 8, 1**.

For **copying and pasting** source code, it is recommended to set **transmission pauses** in your terminal:

1. Delay **per byte**: **~20 ms**
2. Delay **after LF/CR**: **~200 ms**

Since there is **no FIFO buffer**, fast transfers lead to **timing errors**.

Software description

After the **reset**, the 6502 reads the **reset vector** from **\$FFFC/\$FFFD**. The routine initializes the display and serial port and switches to the **monitor**. By default, the ticker text "6502 badge for VCF" / "6502 50th Birthday 1975-2025" runs. The 7 discrete LEDs display a **Knight Rider pattern**. While waiting for serial input, the display is continuously updated.

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6502 Badge – Monitor & Reference Manual

Entering ? + Return displays the help.

Monitor commands

Basic format of a command line:

SSSS.EEEE C <Return>

- SSSS/EEEE = Start/end address (separated by a period)
- C = **Command symbol** (not alphanumeric)
- <Return> = Enter key
- **No spaces** between elements (e.g., 1234.5678L<Return>)

Addresses can have 2–4 hex digits. If more are entered, the **last 4** count. Shorter entries are padded with 0 on the left. Many commands remember the **last** address **used** and **increment** automatically.

?<Return> displays a quick help guide:

Commands are:

Syntax = {} required, [] optional, HHHH hex address, DD hex data

[HHHH][HHHH]{Return}	- Hex dump address(es) (up to 16 without address)
[HHHH]{.HHHH}{Return}	- Hex dump range (16 per line)
[HHHH]{:DD}[DD]{Return}	- Change data bytes
[HHHH]{G}{Return}	- Run program (RTS returns to monitor)
{HHHH.HHHH>HHHHI}{Return}	- Insert range (2.) down into 1.→3.
[HHHH]{L}{Return}	- Disassemble 20 lines
[HHHH]{.HHHH}{L}{Return}	- Disassemble range
{HHHH.HHHH>HHHHM}{Return}	- Move range (1.→2.) after 3.
[HHHH][HHHH]{Q}{Return}	- Text dump address(es)
[HHHH]{.HHHH}{Q}{Return}	- Text dump range (16/line)
{S}[up to 32 text characters]{Return}	- Set LED message
[HHHH]{U}{Return}	- Upload (PC→SBC, XMODEM/CRC)
[HHHH.HHHH]{X}{Return}	- Download (SBC→PC, XMODEM/CRC)
{V}{Return}	- Monitor version
{P*}{Return}	- Protected shutdown
{!}{Return}	- Mini assembler
{@}{Return}	- EhBASIC cold start
{#}{Return}	- EhBASIC warm start
{?}{Return}	- Display help

The following sections explain the most important commands: **Hex Dump**, **Edit Memory**, **Move/Insert**, **Execute**, **Disassembler**, **Text Dump**, **XMODEM**, **Version**, **Power Down**, **Mini Assembler**. (Details as in the original, abridged and linguistically revised.)

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EhBASIC: Cold Start & Warm Start

Cold start {@}<Return> initializes RAM and sets the upper limit ("top of memory"). Input in decimal:

- 2048 → 2 KB RAM
- 32768 → 32 KB RAM

Example (2 KB RAM):

> @

```
Memory size ? 2048
1023 bytes free
Enhanced BASIC 2.22
```

Ready

The first 1 KB of RAM is used by the monitor and display, leaving a total of ≈ 1 KB free. Use SYS<Return> to return from the EhBASIC prompt to the monitor.

Warm start #<Return> then returns to EhBASIC without reinitialization (program in RAM is retained).

EhBASIC quick overview & keywords

EhBASIC by **Lee Davison** (adapted here with kind permission). Manual:

<http://www.sunrise-ev.com/photos/6502/EhBASIC-manual.pdf>.

Keywords (excerpt):

```
ABS AND ASC ATN BIN$ BITCLR BITSET BITTST CALL CHR$ CLEAR CONT COS DATA DEC
DEEK DEF DIM DO DOKE ELSE END EOR EXP FN FOR FRE GET GOSUB GOTO HEX$ IF INC
INPUT INT IRQ LCASE$ LEFT$ LEN LET LIST LOAD LOG LOOP MAX MID$ MIN NEW NEXT
NMI NOT NULL OFF ON OR PEEK PI POKE POS PRINT READ REM RESTORE RETIRQ
RETNMI RETURN RIGHT$ RND RUN SADD SAVE SGN SIN SPC( SQR STEP STOP STR$ SWAP
SYS TAB( TAN THEN TO TWOPI UCASE$ UNTIL USR VAL VARPTR WAIT WHILE WIDTH + -
* / ^ << >> > = <
```

- Keywords are **uppercase** and **without spaces**.
- Numbers: Integer/Decimal/Float; prefix \$ (Hex) or % (Binary), e.g. \$0A, 96.3, 2.718E-3.
- Variables: numeric, strings\$, arrays (n); strings in "...".

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LED operations

The monitor contains routines for **refreshing the 7-segment display** and the **discrete LEDs** (multiplex scan + scroll). Basic variables include:

```
Lbuff = $02A0 ; LED text buffer (max. 32 bytes)
LDbuff = $02C0 ; Buffer discrete LEDs (max. 32 bits)
Lptr = $E2    ; LED pointer (0..31)
Ldig = $E3    ; Digit counter (0..7; 0 = discrete LEDs)
Lscn = $E4    ; Scan delay (brightness)
Lscl = $E5    ; Scroll delay
Lscnc = $E6   ; Scan counter
Lsclc = $E7   ; Scroll counter
LEDchk1 = $E8 ; Config checksum 1
LEDchk2 = $E9 ; Config checksum 2
```

LEDchk1=\$A5 **and** LEDchk2=\$5A prevent the buffers from being overwritten during reset (otherwise standard text "6502 badge for VCF").

Serial input/output (serial operations)

Monitor subroutines (depending on ROM date; check `v` command):

- **Serial_Output** (\$EAE8 or \$EAE9) – sends 1 byte (A)
- **Serial_Input** (\$EB35 or \$EB39) – blocking read (A)
- **Scan_Input** (\$EB4C or \$EB50) – non-blocking (Carry=0 if nothing is there)

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System memory usage

Zero Page

Range	Usage
\$00-\$13	EhBASIC
\$14-\$31	Free
\$32-\$3F	Monitor variables
\$40-\$5A	free
\$5B-\$DF	EhBASIC
\$E2-\$E9	LED driver
\$EA-\$ED	Serial driver
\$EF-\$FF	EhBASIC

RAM

Range	Usage
\$0100-\$01FF	System stack
\$0200-\$027F	Serial receive ring buffer
\$02A0-\$02DF	LED buffer
\$0300-\$037F	Monitor input buffer
\$0390-\$03FF	EhBASIC input buffer
\$0400-\$7FFF	User RAM / EhBASIC program (32 KB)

ROM (depending on monitor date)

5-17-17 ROM	2-24-18 ROM	Description
\$C000-\$C1FF	\$C000-\$C1FF	CRC table (XMODEM)
\$C200-\$EA9B	\$C200-\$EA9F	EhBASIC
\$EA9C-\$EB64	\$EAA0-\$EB68	Serial I/O routines
\$EB65-\$FC39	\$EB69-\$FC3D	Monitor
\$FC3A-\$FD93	\$FC3E-\$FD97	LED support
\$FD94-\$FFBE	\$FD98-\$FFC2	XMODEM
\$FFBF-\$FFFF	\$FFC3-\$FFFF	Reset & vectors

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Source code organization & build notes

With the exception of EhBASIC, the software was written by **Daryl Rictor**. Use and adaptation for **non-commercial** purposes is permitted.

Files

- `sbcb.asm` – "Makefile": loads all sources in the correct order/address.
- `basic.asm` – EhBASIC source.
- `basldsv.asm` – Patch for load/save via XMODEM.
- `sbcbOS.asm` – Monitor including disassembler & mini-assembler.
- `serial.asm` – Software serial driver.
- `LEDdrive.asm` – LED driver (refresh, scroll, text functions).
- `font.asm` – LED font (ASCII mapping to latch pins).
- `xmodem.asm` – XMODEM protocol.
- `CRCtable.asm` – CRC lookup table.
- `reset.asm` – Reset/initialization; ROM vectors `$FFFA-$FFFF`; BRK handler (`$00`) → Reset & Monitor.

Assemble (6502 macro assembler & simulator, `6502.exe`)

1. Start program
2. Open `sbcb.asm` (File → Open → `sbcb.asm`)
3. Set options:
 - Assembler → **Extra byte after BRK: deactivate**
 - General → Select **65C02, 6501**
4. **F7** (Assemble)
5. Save object (File → Save Code): Intel-Hex, S-Record, or Binary
 - **Start address:** `0xC000` (16 KB ROM) or `0x8000` (32 KB ROM)
 - **End address:** `0xFFFF`

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Downloads/Firmware/Manuals

1. Customized firmware (32 KB, 27C256, **7× LEDs**) → https://github.com/FraEgg/The-MOS-6502-50th-Anniversary-Computer-Badge/blob/2.6/downloads/BadgeOS_32k_50th.rom (BIN, 32 KB)
2. Original **BadgeOS (firmware/ROM)** → <https://github.com/FraEgg/The-MOS-6502-50th-Anniversary-Computer-Badge/blob/2.6/downloads/BadgeOS.zip> (ZIP, ~0.8 MB)
3. **Badge Manual** (Lee Hart, Daryl Rictor, Josh Bensadon) → <downloads/badge-manual.pdf> (PDF, ~1.6 MB)
4. **EhBASIC Manual** → <https://github.com/FraEgg/The-MOS-6502-50th-Anniversary-Computer-Badge/blob/2.6/downloads/EhBASIC-manual.pdf> (PDF, ~0.5 MB)
5. **Circuit diagram** → https://github.com/FraEgg/The-MOS-6502-50th-Anniversary-Computer-Badge/blob/2.6/downloads/Schematic_v2.6.pdf (PDF, ~0.2 MB)
6. **PCB layout (Gerber)** → <https://github.com/FraEgg/The-MOS-6502-50th-Anniversary-Computer-Badge/tree/2.6/gerber>
7. **BOM files** → <https://github.com/FraEgg/The-MOS-6502-50th-Anniversary-Computer-Badge/tree/2.6/bom>
8. Driver for **CH340 USB-C ↔ TTL** → <https://www.arduined.eu/tag/ch340/>
9. Additional downloads → <https://github.com/FraEgg/The-MOS-6502-50th-Anniversary-Computer-Badge/tree/2.6/downloads>

Order

The project is available as a "Shared Project" at **PCBWay**:

https://www.pcbway.com/project/shareproject/The_MOS_6502_50th_Anniversary_Computer_Badge_1975_2025_439a8755.html

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Troubleshooting

1. **Power LED is lit, but display flickers.**
Check **U2 pin 8**: is the voltage stable at **4.5–5 V**? If not, check the power source. If the 5 V step-up receives an input voltage of **< 1.2 V**, it breaks down and pulses.
2. **Some LEDs are dark or permanently lit (including 7-segment).**
Check **U6/U7 (74HC273)** for solder bridges/cold solder joints.
3. **Power LED is lit, but no other functions.**
 - 1) Check **U3 (74HC139)** (short circuits/contact).
 - 2) **U1 (EPROM)** and **U2 (CPU)** correctly inserted/program present?
 - 3) **Is U4 (SRAM)** correct and making contact?
 - 4) **Is X1 (oscillator)** correctly fitted?
4. **No USB connection (Windows).**
Install **CH340 driver** (see **Downloads**).
5. **CH340 LED lights up in battery mode even when OFF.**
Normal: The CH340 module is powered by the battery. This keeps **IRQ high**, even without a USB cable. Otherwise, the CPU ends up in an IRQ loop and does not start.

Disclaimer

This is a **non-commercial hobby project**. Replication and use of the **MOS 6502 – 50th Anniversary Computer Badge (1975–2025)** is **at your own risk**. There is **no guarantee** of functionality, safety, completeness, or accuracy. The creator (Frank Eggen) is **not** liable for any damage or consequential damage resulting from replication or use.

Donations

I have invested many hours in this project. If you would like to support my work, I would appreciate a contribution to the coffee fund. **Thank you very much!**

PayPal: https://www.paypal.com/donate/?cmd=_s-xclick&hosted_button_id=Q8HXKYARXKT4L&ssrt=1714757590172

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2. **Chuck Peddle** and the team at **MOS** (including Terry Holdt, Wil Mathys, Rod Orgill, Harry Bawcom, Sydney Anne Holt, Walt Eisenhower, and John Paivinen) for the 6502 → team6502.org.
3. **Bill Mensch** (WDC) for documentation and further development of the 6502 → westerndesigncenter.com.
4. **Ben Eater** for "Build a 6502 computer" → eater.net/6502.